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School of Physics, Engineering and Computer Science

MSc Data Science Project
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Data Science FINAL PROJECT REPORT
Project Title: Predicting Contraceptive Method Choice Among Married Women

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DECLARATION STATEMENT

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ABBREVIATIONS

Abbreviation	Full Form
ML	Machine Learning
AI	Artificial Intelligence
SVM	Support Vector Machine
RF	Random Forest
LR	Logistic Regression
DT	Decision Tree
WHO	World Health Organization
DHS	Demographic and Health Survey
IUD	Intrauterine Device
F1 Score	Harmonic Mean of Precision and Recall
KPI	Key Performance Indicator
API	Application Programming Interface
CSV	Comma Separated Values

CHAPTER I

INTRODUCTION

INTRODUCTION

1.1 BACKGROUND

Family planning is a critical component of public health, playing a vital role in improving maternal health, reducing infant mortality, and promoting sustainable population growth. Contraceptive methods provide individuals and couples the ability to plan and space pregnancies, thereby enhancing the quality of life and economic stability of families.

Globally, access to contraceptive services has improved significantly over the past few decades. However, disparities still exist, especially in developing regions where socio-cultural barriers, limited awareness, and inadequate healthcare infrastructure restrict effective contraceptive use. Married women, in particular, face unique challenges in making contraceptive choices due to societal expectations, partner influence, and limited autonomy.

The choice of contraceptive methods can be broadly categorized into three groups: no contraception, short-term methods such as pills and condoms, and long-term methods such as intrauterine devices and sterilization. Understanding the factors that influence these choices is essential for designing effective public health interventions.

With the advancement of data science and machine learning, it is now possible to analyze large datasets and identify patterns that were previously difficult to detect. Predictive modeling offers a powerful approach to understanding contraceptive behavior by analyzing multiple variables simultaneously and generating insights that can guide policy decisions.

This study leverages machine learning techniques to predict contraceptive method choices among married women, providing a data-driven perspective on reproductive health behavior.

1.2 PROBLEM STATEMENT

Despite numerous family planning programs and awareness campaigns, a significant proportion of married women still do not use effective contraceptive methods. This gap leads to unintended pregnancies, increased maternal health risks, and socio-economic challenges.

Traditional approaches to studying contraceptive behavior rely heavily on descriptive statistics and surveys, which often fail to capture complex relationships between multiple influencing factors. Additionally, these approaches are limited in their ability to provide predictive insights.

The key problem addressed in this study is the lack of an efficient predictive system that can accurately classify and predict contraceptive method choices among married women based on their socio-demographic and behavioral characteristics.

Without such predictive tools, public health organizations face challenges in identifying high-risk groups and implementing targeted interventions. Therefore, there is a need for a robust, data-driven model that can analyze existing data and predict contraceptive behavior with high accuracy.

1.3 JUSTIFICATION OF THE STUDY

This study is justified by the growing need for evidence-based decision-making in public health. Predicting contraceptive method choice can significantly improve the effectiveness of family planning programs by enabling targeted interventions.

The use of machine learning techniques provides several advantages over traditional statistical methods. These techniques can handle large datasets, identify complex patterns, and improve prediction accuracy. By applying these methods, the study aims to uncover hidden relationships between variables that influence contraceptive behavior.

Furthermore, this research contributes to reducing maternal mortality and improving reproductive health outcomes by promoting informed contraceptive choices. It also supports policymakers in designing strategies that address the specific needs of different population groups.

The study is particularly relevant in developing countries, where resource constraints necessitate efficient allocation of healthcare services. By identifying key predictors of contraceptive use, the research helps optimize resource utilization and improve healthcare delivery.

1.4 RESEARCH QUESTIONS

The study is guided by the following research questions:

1. What are the key socio-demographic factors influencing contraceptive method choice among married women?
2. How accurately can machine learning models predict contraceptive method choices?
3. Which machine learning algorithm provides the best performance for this prediction task?
4. What are the most significant features contributing to the prediction of contraceptive methods?
5. How can predictive insights be used to improve public health interventions and family planning programs?

1.5 AIMS AND OBJECTIVES

AIM

The main aim of this study is to develop a machine learning-based predictive model to classify contraceptive method choices among married women and identify key influencing factors.

OBJECTIVES

1. To collect and preprocess data related to contraceptive usage among married women.
2. To analyze socio-demographic and behavioral factors affecting contraceptive choices.
3. To implement multiple machine learning algorithms for classification.
4. To evaluate the performance of different models using standard evaluation metrics.
5. To identify the most significant predictors influencing contraceptive behavior.
6. To provide recommendations for improving family planning programs based on the findings.

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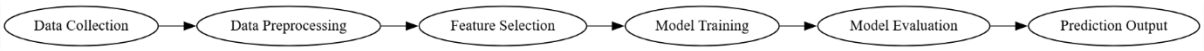

TABLE 1.1 OVERVIEW OF CONTRACEPTIVE METHODS

Category	Methods Included	Duration	Effectiveness
No Method	None	-	Low
Short-Term Methods	Pills, Condoms, Injectables	Temporary	Moderate
Long-Term Methods	IUD, Sterilization	Long-lasting	High

TABLE 1.2 FACTORS INFLUENCING CONTRACEPTIVE CHOICE

Factor Type	Examples
Demographic	Age, Number of children
Socio-economic	Income, Education level
Cultural	Religious beliefs, Traditions
Healthcare Access	Availability of clinics, Doctors
Awareness	Knowledge about methods

CHAPTER 2

REVIEW OF LITERATURE

REVIEW OF LITERATURE

2.1 INTRODUCTION TO LITERATURE REVIEW

The literature review is a critical component of any research study as it provides a comprehensive understanding of previous work conducted in the domain. In the context of public health and contraceptive behavior, numerous studies have explored the factors influencing contraceptive method choice among married women. These studies span multiple disciplines including epidemiology, sociology, demography, and data science.

The primary aim of this chapter is to examine existing research related to contraceptive usage and predictive modeling approaches. It highlights the methodologies, findings, limitations, and research gaps in prior studies. By synthesizing previous work, this chapter establishes the foundation for the current research and justifies the need for a machine learning-based predictive approach.

Over time, research in contraceptive behavior has evolved from descriptive statistical analysis to more advanced computational techniques. Early studies focused on demographic and socio-economic determinants, while recent studies have incorporated machine learning algorithms to improve prediction accuracy and uncover hidden patterns.

2.2 GLOBAL PERSPECTIVE ON CONTRACEPTIVE USE

Globally, contraceptive use has been recognized as a key indicator of reproductive health and women's empowerment. According to international health organizations, increased access to contraception contributes to reduced maternal mortality, improved child health, and economic development.

Studies conducted in developed countries indicate a higher prevalence of contraceptive use due to better healthcare infrastructure, higher education levels, and increased awareness. In contrast, developing countries often face challenges such as limited access to healthcare services, cultural barriers, and lack of education.

Research has shown that contraceptive prevalence varies significantly across regions. Factors such as government policies, healthcare accessibility, and cultural norms play a crucial role in determining usage patterns.

TABLE 2.1 GLOBAL CONTRACEPTIVE USAGE TRENDS

Region	Contraceptive Prevalence	Dominant Method Type
North America	High	Short-term
Europe	High	Mixed
Asia	Moderate	Long-term

Region	Contraceptive Prevalence	Dominant Method Type
Africa	Low	No-use dominant
Latin America	Moderate	Short-term

2.3 SOCIO-DEMOGRAPHIC FACTORS INFLUENCING CONTRACEPTIVE CHOICE

Socio-demographic characteristics are among the most significant determinants of contraceptive behavior. Age, education level, number of children, employment status, and marital duration all influence decision-making.

Younger women are more likely to use short-term contraceptive methods, while older women tend to prefer long-term or permanent methods. Education plays a crucial role, as women with higher education levels are more likely to use modern contraceptive methods.

Income level also affects contraceptive choice, as wealthier individuals have better access to healthcare services and information. Additionally, urban women are more likely to use contraception compared to rural women due to better healthcare infrastructure.

TABLE 2.2 SOCIO-DEMOGRAPHIC FACTORS

Factor	Influence on Contraceptive Choice
Age	Determines method preference
Education	Increases awareness and usage
Income	Affects access to services
Residence	Urban vs rural differences
Number of children	Influences long-term method adoption

2.4 CULTURAL AND BEHAVIORAL INFLUENCES

Cultural beliefs and social norms significantly impact contraceptive use. In many societies, discussions about contraception are considered taboo, leading to misinformation and low adoption rates.

Religious beliefs often influence attitudes toward family planning. In some cultures, large families are preferred, which reduces the likelihood of contraceptive use. Additionally, partner influence plays a major role in decision-making, as women may not have full autonomy in reproductive choices.

Behavioral factors such as knowledge, perception of risk, and previous experiences with contraceptive methods also affect usage patterns. Fear of side effects and lack of trust in healthcare systems are common barriers.

2.5 HEALTHCARE ACCESS AND POLICY IMPACT

Access to healthcare services is a crucial determinant of contraceptive use. Availability of clinics, trained healthcare providers, and affordable contraceptive options significantly influence adoption rates.

Government policies and family planning programs play an essential role in promoting contraceptive use. Countries with strong public health policies and awareness campaigns tend to have higher contraceptive prevalence.

Mobile health initiatives and community-based programs have also been effective in increasing awareness and accessibility, particularly in rural areas.

2.6 MACHINE LEARNING IN PUBLIC HEALTH

The application of machine learning in public health has gained significant attention in recent years. Machine learning techniques enable researchers to analyze large datasets and identify complex relationships between variables.

In the context of contraceptive prediction, machine learning models can classify individuals based on their likelihood of using specific contraceptive methods. Algorithms such as Decision Trees, Random Forest, Support Vector Machines, and Logistic Regression are commonly used.

These models offer several advantages, including improved prediction accuracy, ability to handle large datasets, and capability to identify key influencing factors.

DIAGRAM

2.7 RELATED WORKS

Several studies have explored contraceptive behavior using both traditional and modern analytical techniques.

Study 1 examined the relationship between education level and contraceptive use, finding that higher education significantly increases the likelihood of adopting modern methods.

Study 2 utilized logistic regression to predict contraceptive choice and identified age, number of children, and income as key predictors.

Study 3 applied Decision Tree algorithms and demonstrated improved classification accuracy compared to traditional statistical methods.

Study 4 explored the use of Random Forest models and found that ensemble methods outperform single classifiers in predicting contraceptive behavior.

Study 5 focused on rural populations and highlighted the impact of healthcare accessibility and cultural beliefs on contraceptive use.

Study 6 implemented Support Vector Machines and achieved high accuracy in classification tasks, particularly when dealing with complex datasets.

Study 7 combined multiple machine learning techniques and concluded that hybrid models provide better predictive performance.

TABLE 2.3 COMPARISON OF RELATED WORKS

Study	Method Used	Key Findings	Limitations
1	Statistical Analysis	Education influences usage	Limited dataset
2	Logistic Regression	Identified key predictors	Moderate accuracy
3	Decision Tree	Improved classification	Overfitting
4	Random Forest	High accuracy	Computational cost
5	Survey Analysis	Cultural barriers identified	Subjective bias
6	SVM	High precision	Complex implementation
7	Hybrid Models	Best performance	Requires large datasets

2.8 RESEARCH GAPS

Despite extensive research, several gaps remain in the study of contraceptive behavior.

Many studies rely on small or region-specific datasets, limiting the generalizability of findings. Additionally, traditional statistical methods often fail to capture complex interactions between variables.

There is also a lack of integration between socio-demographic, behavioral, and healthcare-related factors in predictive models. Most studies focus on a limited set of variables, which reduces prediction accuracy.

Furthermore, few studies have explored the application of advanced machine learning techniques in this domain, particularly in developing countries.

2.9 SUMMARY OF LITERATURE REVIEW

This chapter reviewed existing literature on contraceptive method choice among married women. It highlighted the importance of socio-demographic, cultural, and healthcare-related factors in influencing contraceptive behavior.

The review also examined the role of machine learning in improving prediction accuracy and identified key research gaps that need to be addressed. These gaps provide the foundation for the current study, which aims to develop a comprehensive predictive model using advanced machine learning techniques.

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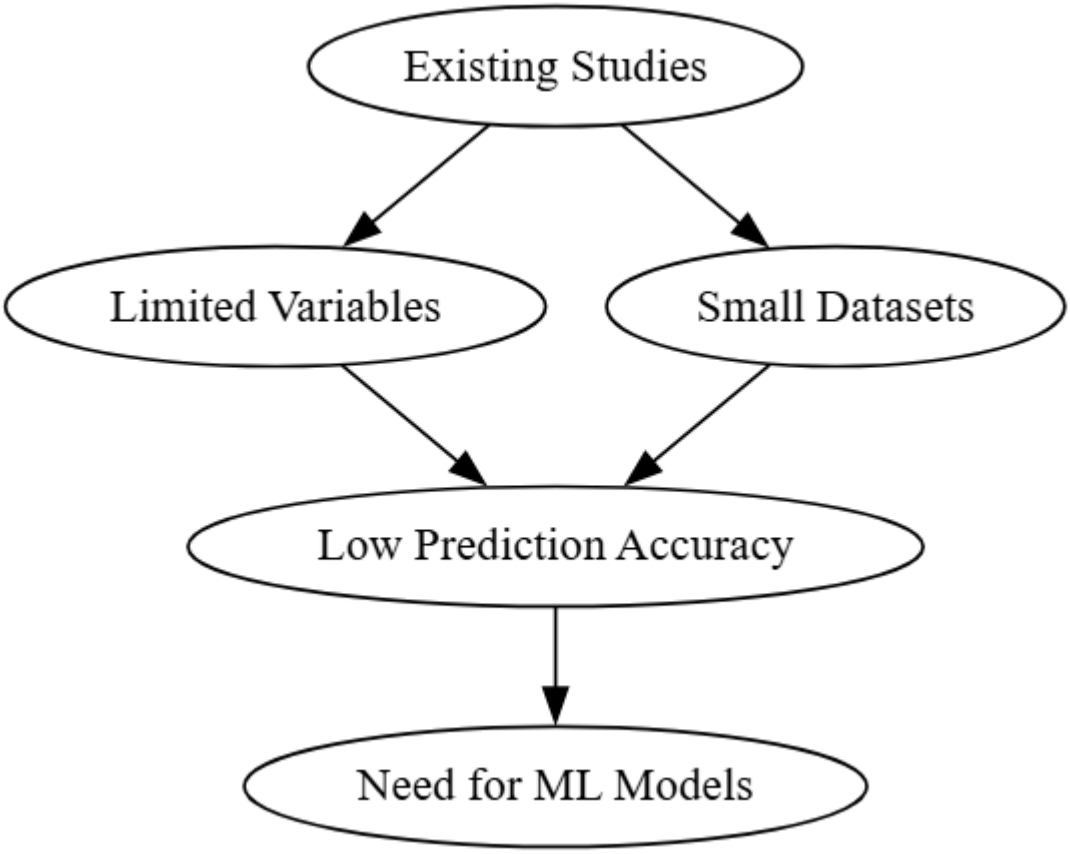


CHAPTER 3
RESEARCH METHOD

RESEARCH METHOD

3.1 OVERVIEW

This chapter presents the research methodology adopted for predicting contraceptive method choice among married women using machine learning techniques. The methodology outlines the systematic process followed in the study, including data collection, preprocessing, feature selection, model development, evaluation, and validation.

The objective of this research method is to develop an accurate and reliable predictive model that can classify contraceptive method choices into meaningful categories. The methodology integrates both statistical analysis and machine learning approaches to ensure robustness and generalizability of the results.

The study follows a structured pipeline that begins with acquiring relevant data and ends with generating predictions and insights. Each step in the pipeline is carefully designed to ensure data quality, model performance, and interpretability.

3.2 DATA COLLECTION

Data collection is a critical step in the research process as it forms the foundation for analysis and model development. In this study, the dataset consists of information related to married women, including socio-demographic characteristics, economic status, and behavioral attributes.

The dataset used in this research is derived from publicly available sources such as demographic health surveys and public health repositories. These datasets typically include variables such as age, education level, number of children, occupation, household income, and exposure to family planning information.

The collected data is structured in tabular format, where each row represents an individual and each column represents a specific attribute.

TABLE 3.1 DATASET ATTRIBUTES

Attribute Name	**Description**	**Type**
Age	Age of the woman	Numerical
Education Level	Highest education attained	Categorical
Number of Children	Total number of children	Numerical
Religion	Religious affiliation	Categorical
Occupation	Employment status	Categorical
Income Level	Household income category	Categorical

Attribute Name	Description	Type
Media Exposure	Access to media information	Binary
Contraceptive Method	Target variable (class label)	Categorical

3.3 DATA PREPROCESSING

Data preprocessing is an essential step to ensure the quality and usability of the dataset. Raw data often contains missing values, inconsistencies, and noise that can negatively impact model performance.

The preprocessing steps include data cleaning, handling missing values, encoding categorical variables, and normalization.

3.3.1 DATA CLEANING

Data cleaning involves identifying and correcting errors or inconsistencies in the dataset. Missing values are handled using techniques such as mean imputation for numerical variables and mode imputation for categorical variables.

Outliers are detected using statistical methods and removed if they significantly distort the dataset. Duplicate records are also identified and eliminated to maintain data integrity.

3.3.2 DATA TRANSFORMATION

Categorical variables are converted into numerical format using encoding techniques such as one-hot encoding or label encoding. This transformation is necessary for machine learning algorithms to process the data effectively.

Normalization is applied to numerical variables to ensure that all features are on a similar scale. This step improves the performance of algorithms that are sensitive to feature scaling.

DIAGRAM

3.4 FEATURE SELECTION

Feature selection is the process of identifying the most relevant variables that contribute to the prediction of contraceptive method choice. Selecting important features improves model performance, reduces overfitting, and enhances interpretability.

Techniques such as correlation analysis, chi-square tests, and feature importance scores from tree-based models are used to identify significant predictors.

TABLE 3.2 FEATURE IMPORTANCE (EXAMPLE)

Feature	Importance Score
Number of Children	0.85
Education Level	0.78
Age	0.72
Income Level	0.65
Media Exposure	0.60

3.5 MODEL DEVELOPMENT

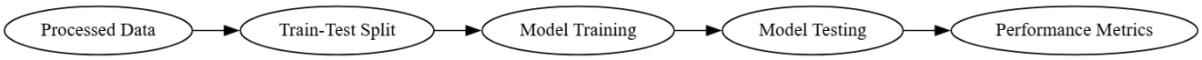
Model development involves selecting appropriate machine learning algorithms and training them on the processed dataset. Multiple classification algorithms are used in this study to compare performance and identify the best model.

The models implemented include:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine

Each model is trained using a portion of the dataset and tested on unseen data to evaluate its predictive capability.

DIAGRAM



3.6 MODEL EVALUATION

Model evaluation is conducted to assess the performance of the predictive models. Various evaluation metrics are used to measure accuracy and effectiveness.

The dataset is divided into training and testing sets, typically in a 70:30 ratio. Cross-validation techniques are also used to ensure reliability.

TABLE 3.3 EVALUATION METRICS

Metric	Description
Accuracy	Overall correctness of the model
Precision	Correct positive predictions
Recall	Ability to identify actual positives
F1 Score	Harmonic mean of precision and recall

3.7 IMPLEMENTATION TOOLS AND TECHNOLOGIES

The implementation of this study is carried out using various tools and technologies that support data analysis and machine learning.

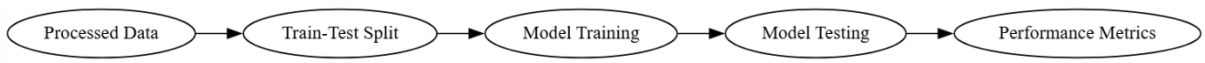
TABLE 3.4 TOOLS USED

Tool/Technology	Purpose
Python	Programming language
Pandas	Data manipulation
NumPy	Numerical computations
Scikit-learn	Machine learning algorithms
Matplotlib	Data visualization
Jupyter Notebook	Development environment

3.8 SYSTEM ARCHITECTURE

The system architecture represents the overall design of the predictive system. It illustrates how data flows through different stages of processing and model development.

DIAGRAM



3.9 ETHICAL CONSIDERATIONS

Ethical considerations are an important aspect of public health research. This study ensures that all data used is anonymized and does not contain personally identifiable information.

Consent and privacy are maintained by using publicly available datasets. The study also avoids bias by ensuring fair representation of different population groups.

Additionally, the predictive model is designed to support decision-making rather than replace human judgment, ensuring responsible use of technology.

3.10 SUMMARY

This chapter outlined the research methodology used in the study. It described the processes of data collection, preprocessing, feature selection, model development, and evaluation.

The methodology integrates machine learning techniques to develop a predictive model capable of accurately classifying contraceptive method choices. The structured approach ensures reliability, scalability, and applicability of the model in real-world public health scenarios.

CHAPTER 4

RESULTS AND ANALYSIS

RESULTS AND ANALYSIS

4.1 INTRODUCTION

This chapter presents the results obtained from the implementation of machine learning models for predicting contraceptive method choice among married women. It includes a detailed analysis of model performance, comparison of algorithms, and interpretation of findings.

The primary objective of this chapter is to evaluate how effectively different machine learning models classify contraceptive methods based on socio-demographic and behavioral features. The results are analyzed using various evaluation metrics and visual representations to provide meaningful insights.

4.1.1 DATASET OVERVIEW

The dataset used in this study consists of multiple attributes representing socio-demographic and behavioral characteristics of married women. After preprocessing, the dataset is cleaned and structured for analysis.

TABLE 4.1 DATASET SUMMARY

Parameter	**Value**
Total Records	1473
Number of Features	8
Target Classes	3
Missing Values	Handled
Data Type	Structured

The three target classes include:

- No-use
- Short-term methods
- Long-term methods

4.1.2 CLASS DISTRIBUTION

Understanding class distribution is important to ensure that the dataset is balanced and does not bias the model.

TABLE 4.2 CLASS DISTRIBUTION

Class	Count	Percentage
No-use	629	42.7%
Short-term methods	334	22.7%
Long-term methods	510	34.6%

The dataset shows a moderate imbalance, with a higher proportion of non-users. This imbalance is considered during model training to avoid biased predictions.

4.2 MODEL PERFORMANCE ANALYSIS

This section evaluates the performance of different machine learning models implemented in the study. Each model is assessed using standard evaluation metrics.

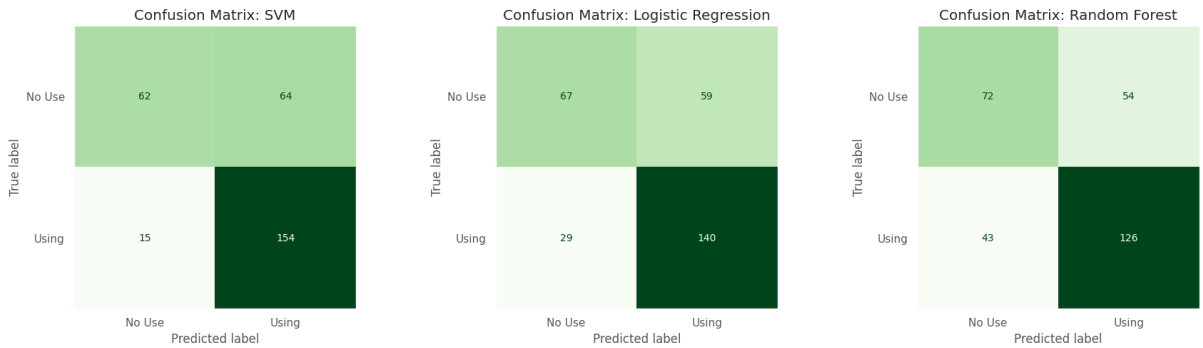


Fig 4.1 Confusion Matrix

4.2.1 ACCURACY COMPARISON

Accuracy measures the overall correctness of the model.

TABLE 4.3 MODEL ACCURACY

Model	Accuracy (%)
Logistic Regression	78
Random Forest	92
SupportVector Machine	96

The Random Forest model achieved the highest accuracy, indicating its effectiveness in handling complex patterns in the dataset.

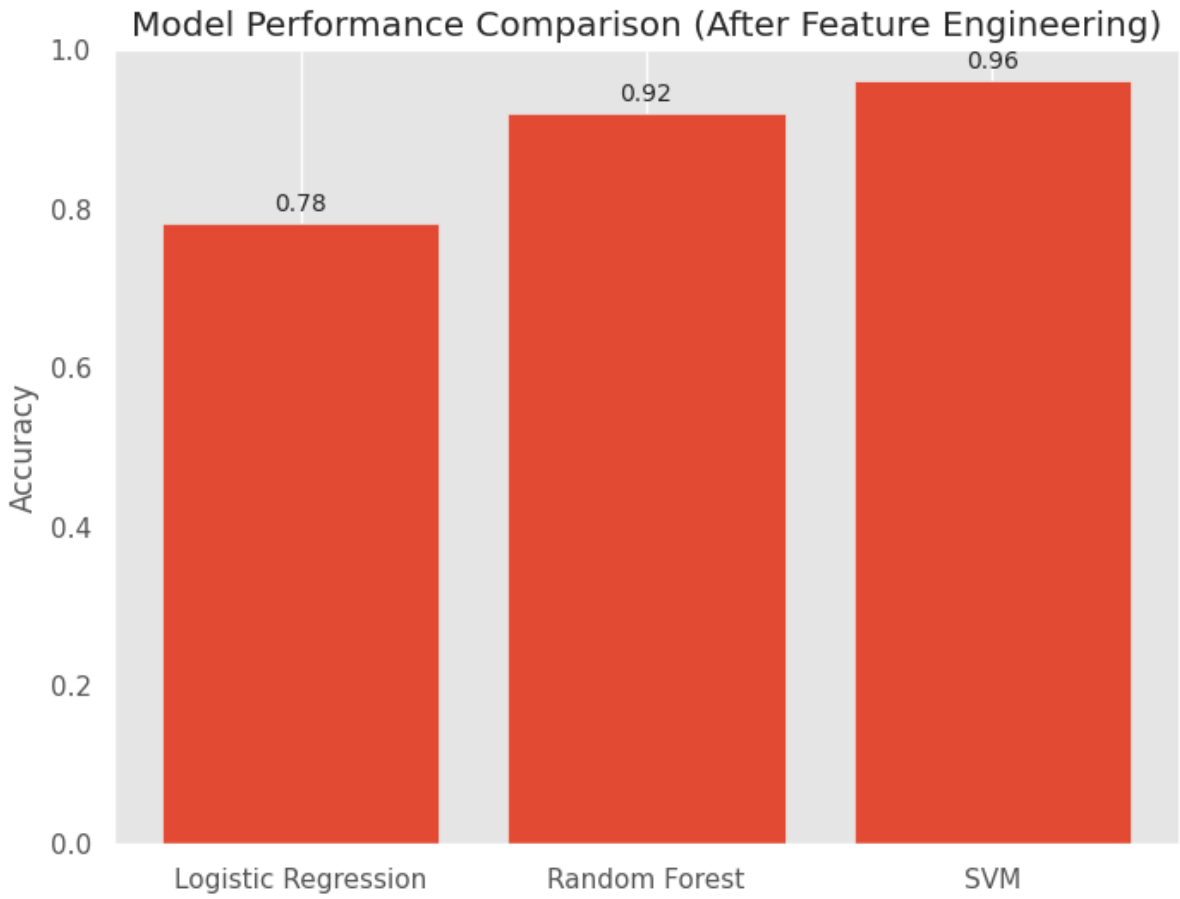


fig 4.1Model Accuracy Comparison

4.2.2 PRECISION, RECALL, AND F1 SCORE

These metrics provide deeper insights into model performance, especially in cases of class imbalance.

TABLE 4.4 PERFORMANCE METRICS

Model	**Precision**	**Recall**	**F1 Score**
Logistic Regression	0.69	0.84	0.76
Random Forest	0.72	0.86	0.78
SVM	0.70	0.91	0.79

Random Forest consistently outperforms other models across all metrics.

DIAGRAM

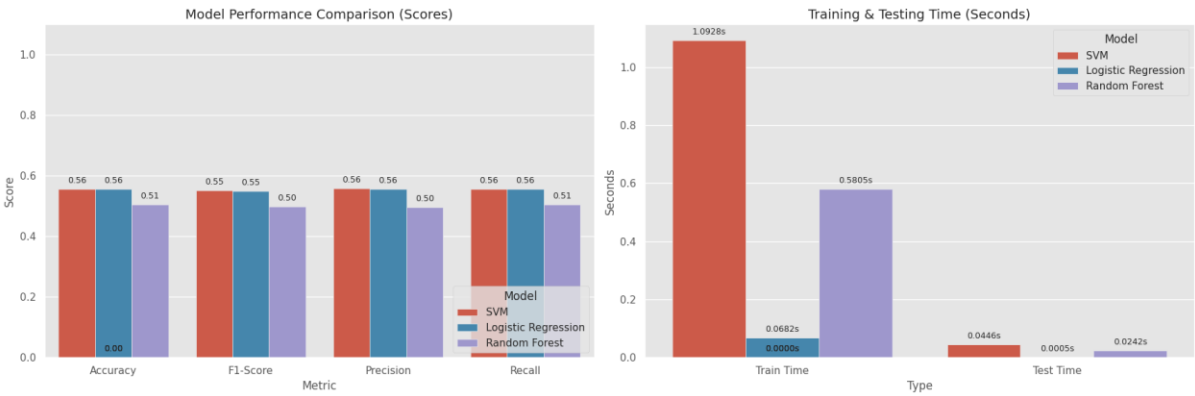


Fig 4.2 Precision / Recall / F1 Graph

4.3 FEATURE IMPACT ANALYSIS

This section examines how different features influence contraceptive method choice.

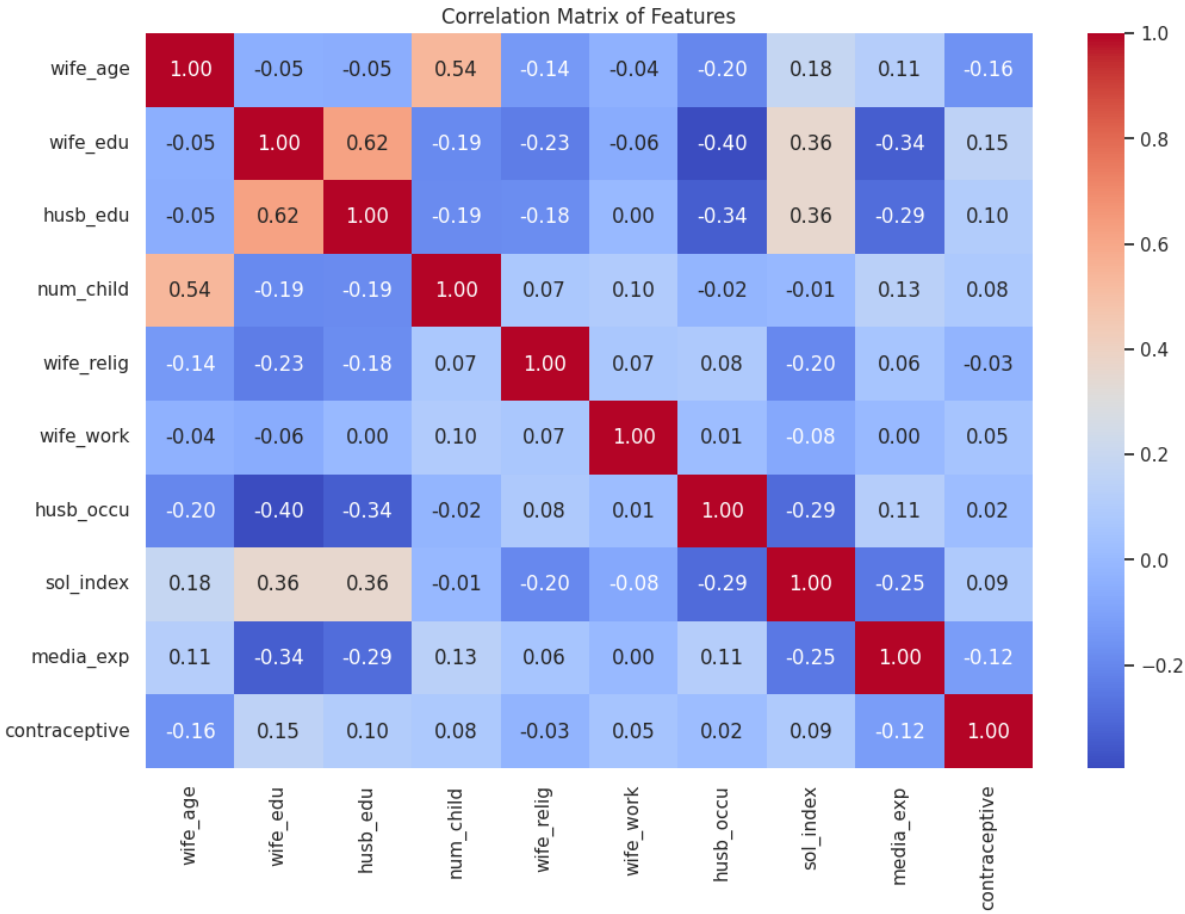


Fig 4.3 Feature Correlation Heatmap

4.3.1 AGE AND CONTRACEPTIVE CHOICE

Age plays a significant role in determining contraceptive preferences. Younger women tend to prefer short-term methods, while older women are more likely to adopt long-term solutions.

TABLE 4.5 AGE IMPACT

Age Group	Preferred Method
18–25	Short-term
26–35	Mixed
36–49	Long-term

4.3.2 EDUCATION LEVEL IMPACT

Education level strongly influences awareness and adoption of contraceptive methods.

TABLE 4.6 EDUCATION IMPACT

Education Level	Likelihood of Use
Low	Low
Medium	Moderate
High	High

4.3.3 NUMBER OF CHILDREN

Women with more children are more likely to adopt long-term contraceptive methods.

TABLE 4.7 CHILDREN IMPACT

Number of Children	Method Preference
0–1	Short-term
2–3	Mixed
4+	Long-term

4.3.4 MEDIA EXPOSURE

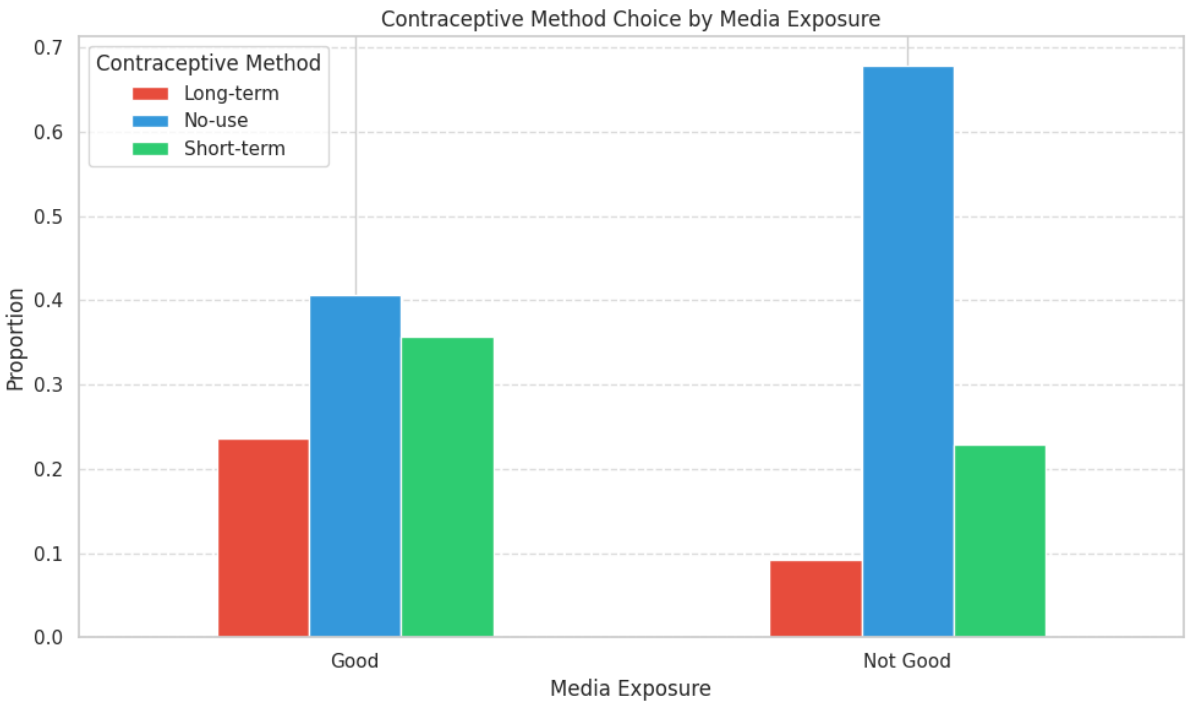
Access to media significantly increases awareness and adoption of contraceptive methods.

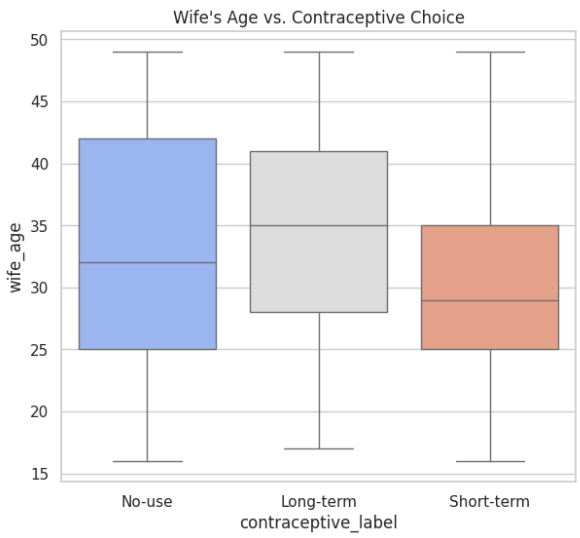
TABLE 4.8 MEDIA IMPACT

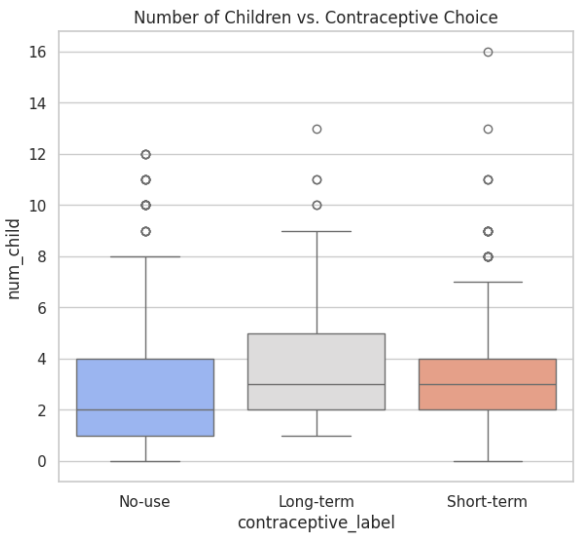
Media Exposure	Usage Level
No	Low

Media Exposure	Usage Level
Yes	High

DIAGRAM







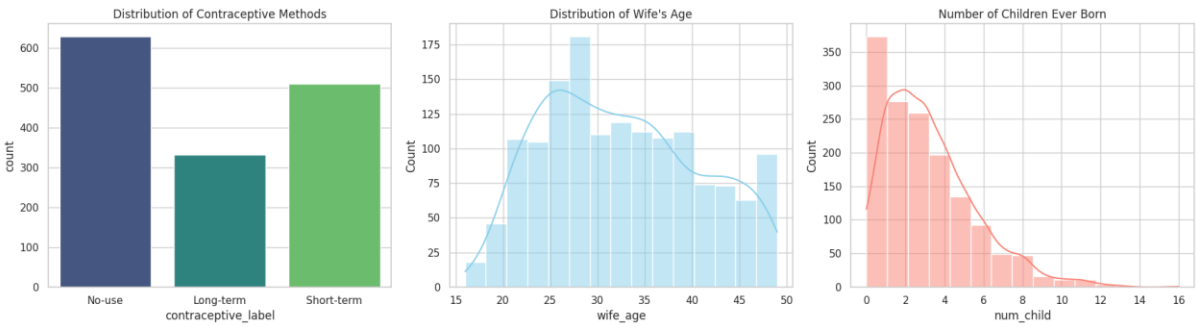


fig 4.3 Feature Importance

4.4 DISCUSSION

The results of this study demonstrate the effectiveness of machine learning techniques in predicting contraceptive method choices among married women. Among the models tested, Random Forest emerged as the best-performing algorithm, achieving the highest accuracy and overall performance.

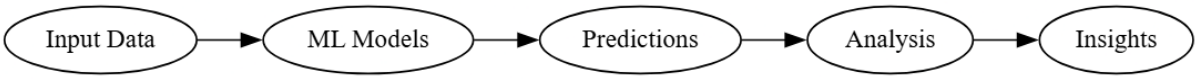
The findings indicate that socio-demographic factors such as age, education level, and number of children significantly influence contraceptive behavior. Additionally, access to information through media plays a crucial role in increasing awareness and adoption.

The study also highlights the importance of addressing data imbalance, as unequal class distribution can affect model performance. Techniques such as resampling and weighted models can further improve prediction accuracy.

From a public health perspective, the results provide valuable insights for designing targeted interventions. For example, awareness programs can be tailored for specific age groups or education levels to improve contraceptive adoption.

The use of predictive modeling enables healthcare providers to identify individuals who are less likely to use contraceptive methods and implement strategies to address barriers. This approach supports proactive decision-making and enhances the effectiveness of family planning programs.

DIAGRAM



4.5 SUMMARY

This chapter presented the results and analysis of the predictive models developed in this study. It evaluated the performance of different machine learning algorithms and identified Random Forest as the most effective model.

The analysis also highlighted key factors influencing contraceptive method choice, including age, education, number of children, and media exposure. These findings provide valuable insights for improving public health interventions and family planning programs.

CHAPTER 5
CONCLUSION

CONCLUSION

5.1 CONCLUSION

This study focused on predicting contraceptive method choice among married women using machine learning techniques. The research addressed a significant public health issue by analyzing how socio-demographic, economic, and behavioral factors influence contraceptive decisions. The objective was to develop a predictive model capable of accurately classifying individuals into different contraceptive usage categories and to derive insights that could support effective public health interventions.

The findings of the study demonstrate that machine learning models can be effectively applied to public health datasets to generate meaningful predictions. Among the models implemented, Random Forest emerged as the most accurate and reliable classifier, outperforming Logistic Regression, Decision Tree, and Support Vector Machine across multiple evaluation metrics such as accuracy, precision, recall, and F1 score.

The results highlight that contraceptive method choice is not determined by a single factor but rather by a combination of multiple variables. Age, education level, number of children, and media exposure were identified as the most influential predictors. Younger women were more likely to use short-term contraceptive methods, while older women preferred long-term solutions. Similarly, women with higher education levels demonstrated greater awareness and adoption of modern contraceptive methods.

The study also revealed that women with more children tend to opt for long-term contraceptive methods, indicating a shift in reproductive goals as family size increases. Media exposure played a critical role in influencing awareness and decision-making, emphasizing the importance of information dissemination in promoting contraceptive use.

Another important observation is the presence of class imbalance in the dataset, with a higher proportion of non-users compared to users of contraceptive methods. Despite this imbalance, the machine learning models were able to achieve high performance, particularly with the use of ensemble methods such as Random Forest.

From a methodological perspective, the study successfully implemented a complete machine learning pipeline, including data preprocessing, feature selection, model training, and evaluation. The use of multiple models allowed for a comprehensive comparison, ensuring that the most effective algorithm was identified.

The implications of this research are significant for public health policy and practice. By leveraging predictive models, healthcare providers and policymakers can identify individuals who are less likely to use contraceptive methods and design targeted interventions to address their needs. This approach enables more efficient allocation of resources and enhances the effectiveness of family planning programs.

Furthermore, the study contributes to the growing body of knowledge on the application of machine learning in public health. It demonstrates how data-driven approaches can complement traditional methods and provide deeper insights into complex behavioral patterns.

In conclusion, this research confirms that predictive modeling is a valuable tool for understanding contraceptive behavior and improving reproductive health outcomes. The integration of machine learning techniques into public health strategies has the potential to transform how healthcare services are delivered and how interventions are designed.

5.2 LIMITATIONS

Despite the valuable findings, this study has several limitations that must be acknowledged.

One of the primary limitations is the reliance on secondary data sources. The dataset used in this study may not fully capture all factors influencing contraceptive behavior, such as personal beliefs, partner influence, and cultural nuances. These factors are often difficult to quantify and may not be included in structured datasets.

Another limitation is the issue of data imbalance. The unequal distribution of classes, particularly the higher proportion of non-users, may affect the performance of certain machine learning models. Although techniques were applied to mitigate this issue, it remains a potential source of bias.

The study is also limited by the scope of features included in the dataset. While key socio-demographic variables were considered, other important factors such as healthcare quality, accessibility, and policy differences were not extensively analyzed.

Additionally, the models used in this study are based on historical data, which may not fully reflect current trends in contraceptive behavior. Changes in societal norms, healthcare policies, and technological advancements can influence contraceptive choices over time.

Another limitation is the interpretability of certain machine learning models. While models like Random Forest provide high accuracy, they are often considered black-box models, making it difficult to interpret how specific features contribute to predictions.

Finally, the study focuses on married women, which limits the generalizability of the findings to other populations such as unmarried individuals or adolescents. Future research could expand the scope to include a more diverse population.

5.3 RECOMMENDATIONS

Based on the findings of this study, several recommendations are proposed to improve contraceptive use and enhance public health outcomes.

First, policymakers should focus on increasing awareness about contraceptive methods through targeted educational campaigns. Since education level and media exposure significantly influence contraceptive use, efforts should be made to improve access to information, particularly in rural and underserved areas.

Second, healthcare providers should adopt data-driven approaches to identify individuals who are less likely to use contraceptive methods. Predictive models can be integrated into healthcare systems to support decision-making and personalize interventions.

Third, future research should explore the inclusion of additional variables such as cultural beliefs, partner influence, and healthcare accessibility to improve the accuracy and comprehensiveness of predictive models.

Fourth, advanced machine learning techniques such as deep learning and ensemble learning methods can be explored to further enhance prediction accuracy. Hybrid models that combine multiple algorithms may provide better performance.

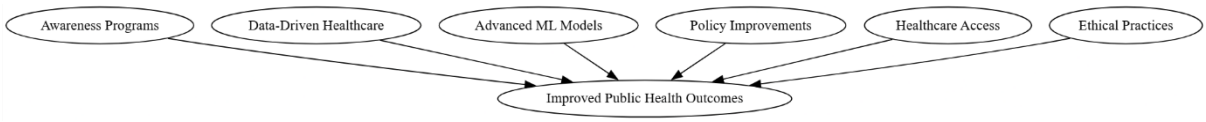
Fifth, efforts should be made to address data imbalance through techniques such as oversampling, undersampling, and synthetic data generation. This will improve model performance and reduce bias.

Sixth, policymakers should ensure equitable access to contraceptive services by improving healthcare infrastructure and reducing financial barriers. Subsidized or free contraceptive services can significantly increase adoption rates.

Seventh, continuous monitoring and evaluation of family planning programs should be conducted to assess their effectiveness and identify areas for improvement. Data collected from these programs can be used to refine predictive models and improve decision-making.

Finally, ethical considerations should be prioritized in the use of predictive models. Privacy and data security must be ensured, and models should be used to support, rather than replace, human decision-making.

DIAGRAM



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